

**UNIVERSITY OF ROME “TOR VERGATA”**

Faculty of Engineering  
M.Sc. in Management Engineering

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**COURSE**

**Management of Innovation and Technology**

**Autodesk Fusion 360**

Project Report

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Project 4 – Generative Design of a Structural Component

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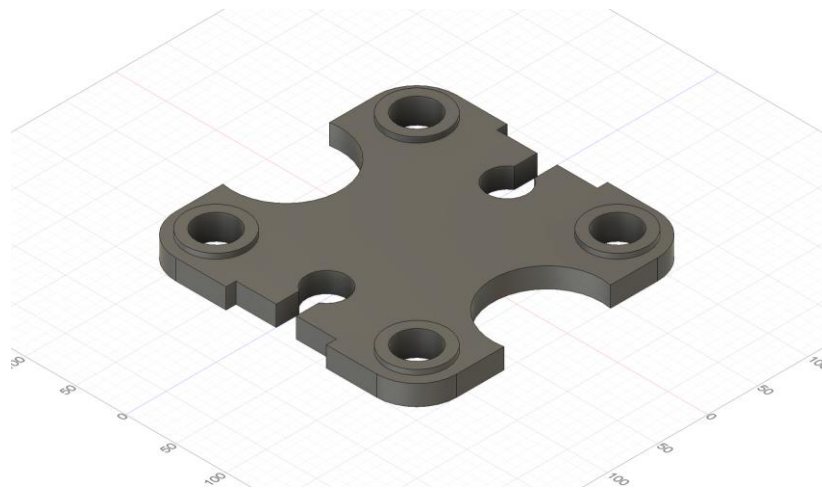
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## 1. Objective

This fourth project applies the **same load case used in Project 3 (topology optimisation)** but follows the opposite logic. In topology optimisation the procedure begins from an existing solid body and *remove* material in order to optimise its shape. In **generative design** no starting body is required: the analyst defines the **conditions to satisfy** (preserved regions, obstacles, constraints, loads), the **objective** (e.g. minimize mass), the **manufacturing technology** and the **candidate materials**, and the software autonomously generates a large number of valid geometries that satisfy them.

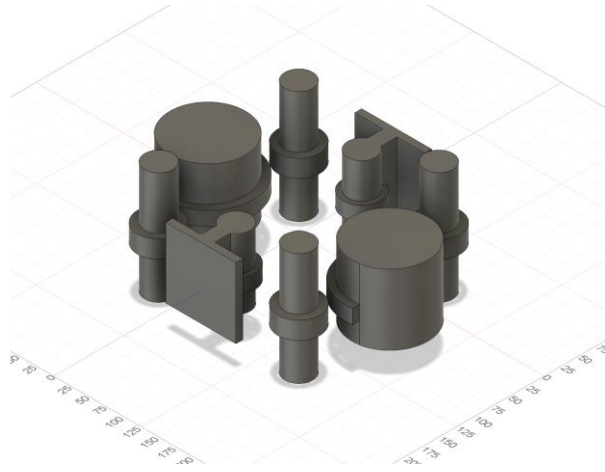
The objective of the study is therefore to adopt the boundary conditions of the Project-3 static structural analysis (a steel component characterised by a given safety factor, von Mises stress and displacement) and to allow generative design to explore a **range of materials and manufacturing methods**, then **compare the generated outcomes against that steel reference**.



## 2. Preparing the geometry: preserve and obstacle bodies

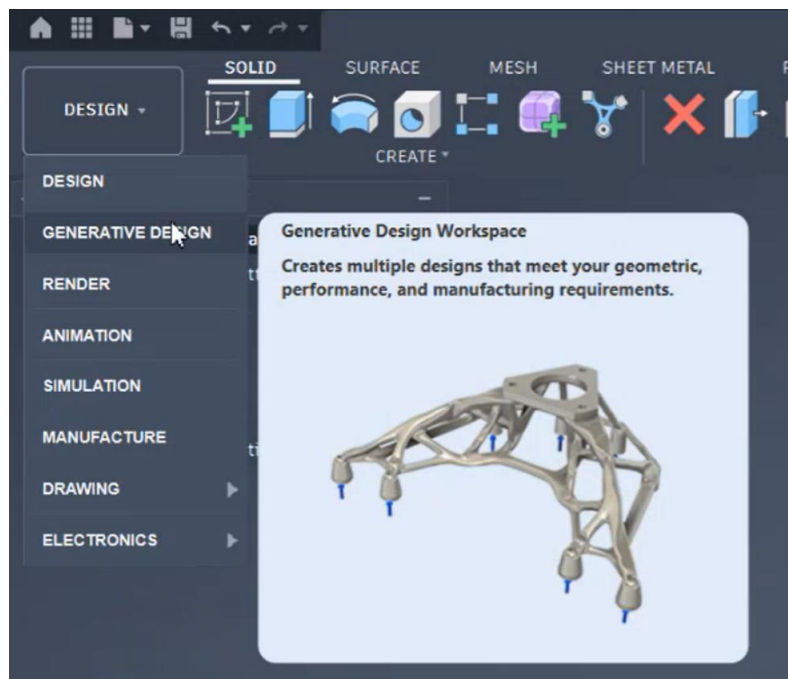
Because generative design operates on regions rather than on a single solid, the geometry is subdivided into smaller bodies, each of which performs a distinct role:

- **Preserve (preservation) geometry:** the regions that **must remain** in every solution. They correspond to the functional zones to which the constraints and loads are applied (the bored bosses and mounting holes). In the present model, these are the four small bodies retained around the holes.
- **Obstacle (limit) geometry:** the regions where the software is **not allowed to add material** (keep-out / interference volumes).



### 3. Creating the Generative Study

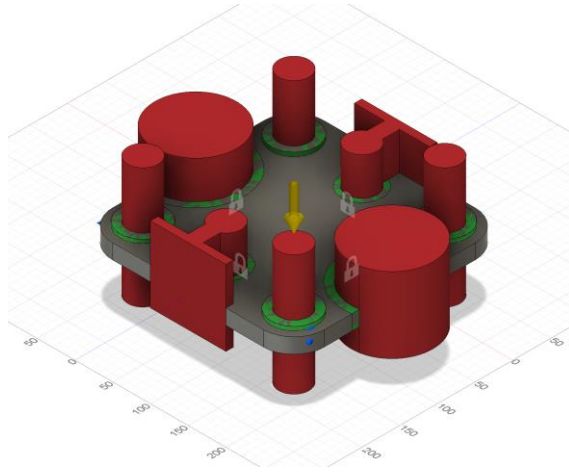
Switch from **DESIGN** to the **GENERATIVE DESIGN** workspace, then **Study** → **New Generative Study**. This creates a *Generative Model* and a *Study* under which all design conditions are defined.



### 4. Design Space — assigning bodies

Under **Design Space**, each body is assigned its role:

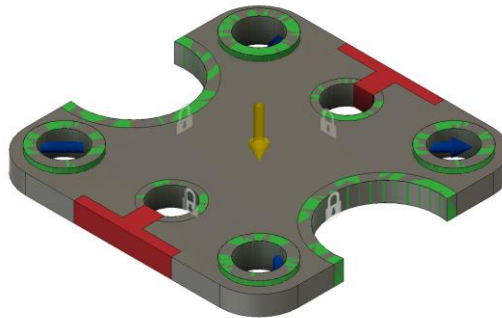
- **Preserve Geometry** → the bodies that must be kept (here, 4 bodies).
- **Obstacle Geometry** → the keep-out bodies (here, 2 bodies).
- *(Optional)* **Symmetry Plane** and **Starting Shape** can also be defined.



#### 4.1 Preserve and obstacle configurations

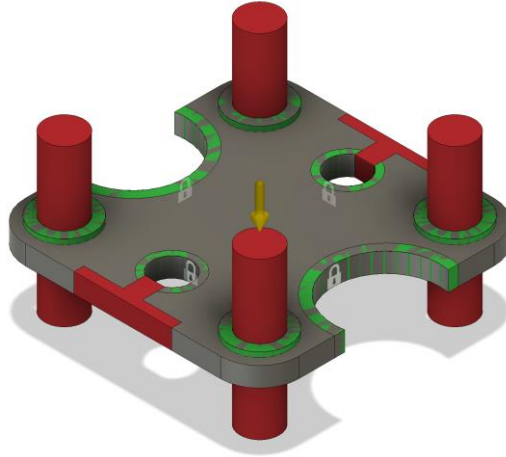
Three distinct preserve and obstacle configurations were defined for the same component, each extending the amount of geometry that the software was required to retain. The three setups are shown below.

In the first configuration, only thin rings of material around the bores of the four corner holes and the two central holes were assigned as **preserve geometry**, while the two lateral T-shaped bodies were defined as **obstacle geometry**. The fixed constraints and the 1000 N force were applied to the corresponding faces.



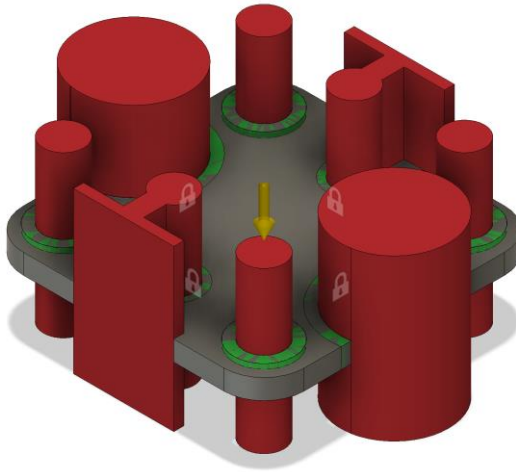
*Configuration 1. preserve rings on the holes; lateral T-bodies as obstacles.*

In the second configuration, tall cylindrical **obstacle bodies** were added along the axes of the corner and central holes. These keep-out volumes prevent the software from filling the bores, so that the holes remain open and functional in every generated outcome.



*Configuration 2. cylindrical obstacle bodies added along the hole axes.*

In the third configuration, the preserved and protected geometry was extended further, adding obstacle bodies around the full set of holes and along the perimeter. This is the most constrained setup, as every functional hole is explicitly protected.



*Configuration 3 — extended preserve and obstacle geometry, adopted in this report.*

The third configuration was adopted for the study presented in this report. Because all of the holes are functionally important, they were all specified as protected regions; this prevents the generative solver from closing or obstructing any of them and, in turn, yields better generated shapes than the less constrained setups. The outcomes discussed in the following sections were obtained using this configuration.

## 5. Design Conditions — constraints and loads

The boundary conditions reuse the Project-3 static structural setup:

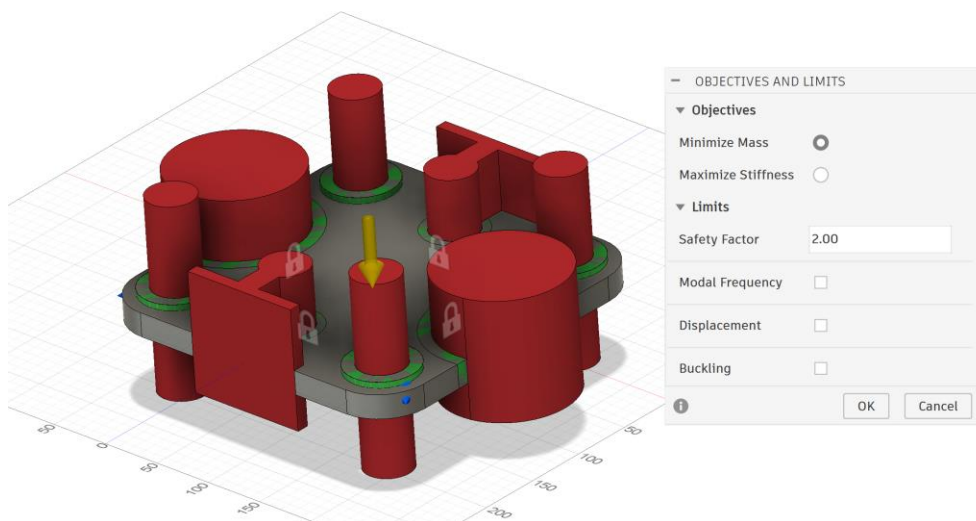
- **Structural constraints** → **Fixed** support applied to the relevant faces (the mounting/fixed faces).
- **Structural loads** → a **Force** of **1000 N** applied to the loaded faces, in the defined direction.



## 6. Objectives and Limits

Under **Design Criteria** → **Objectives and Limits**:

- **Objective:** *Minimize Mass* (alternatively *Maximize Stiffness*).
- **Limit:** **Safety Factor = 2.00**.
- Optional limits (Modal Frequency, Displacement, Buckling) are available for dynamic or stiffness-critical components.

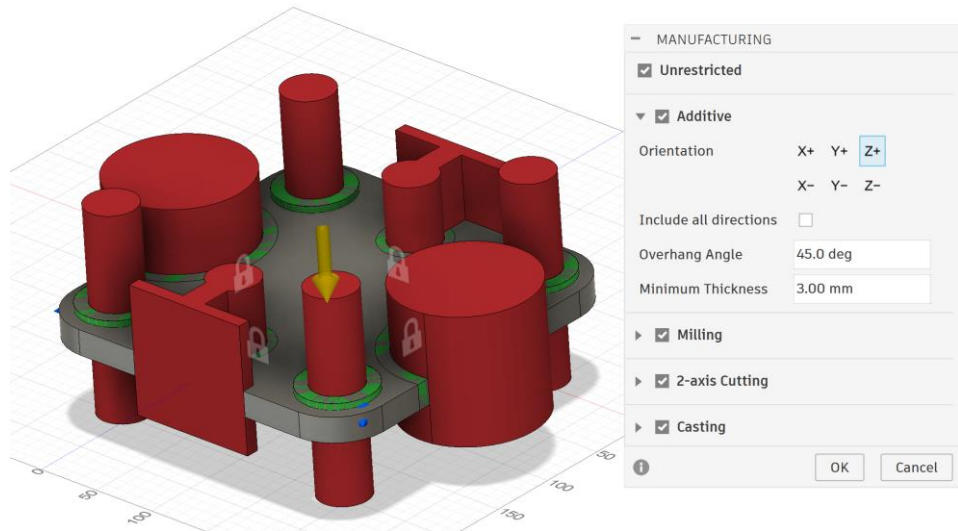


## 7. Manufacturing criteria (technologies)

Several manufacturing methods are enabled so the study explores how the optimal shape changes with the production process:

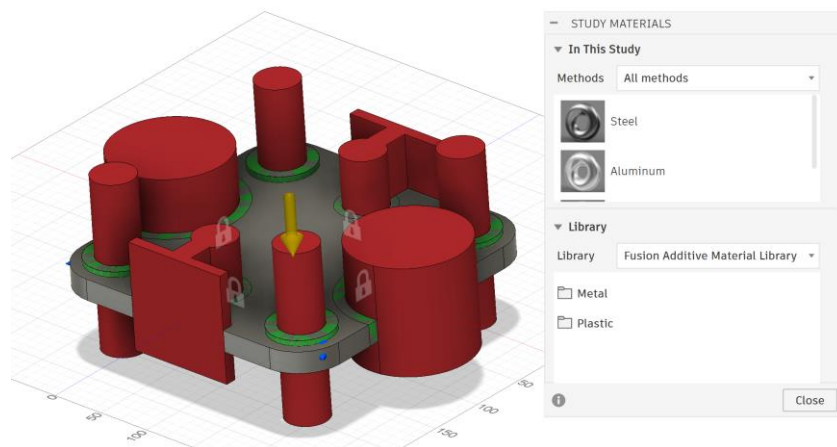
- **Additive** (3D printing)

- **Milling**
- **2-axis cutting** (laser / waterjet / plasma)
- **Casting**
- **Unrestricted** (no manufacturing limit — pure optimum)



## 8. Materials

A range of candidate materials is selected so cost, weight and stiffness can be traded off in a single study. In this project the set spanned **Steel, Aluminium (generic, AlSi10Mg, 6061, A356-T6) and ABS Plastic**.



## 9. Exploring the outcomes

Once the solution has completed, the **Explore** shows all generated geometries. They can be **sorted/filtered** by Mass, Volume, Min Safety Factor, Max von Mises stress or Max displacement, viewed as **thumbnails**, a **property comparison table**, or a **scatter plot** against any chosen manufacturing method or performance metric.

Outcome filters

Processing status

Generative Models

Study

Visual similarity

Manufacturing method

Materials

Objective ranges

Volume (mm<sup>3</sup>)

369,737,469

1.444e+6

Mass (kg)

0.392

11.33

Max von Mises stress (MPa)

0.919

3.427

Min safety factor

5.869

298.283

Max displacement global (mm)

3.086e-4

0.232

| Name                                                                                                                | Preview Features | Recommendation ↓ | Processing status | Generative model   | Material          | Manufacturing method | Visual similarity |
|---------------------------------------------------------------------------------------------------------------------|------------------|------------------|-------------------|--------------------|-------------------|----------------------|-------------------|
|  Study 2 - Structu... - Outcome 13 |                  | 90.022 %         | Converged         | Generative Model 1 | ABS Plastic       | 2 axis cutting       | Group 3           |
|  Study 2 - Structu... - Outcome 19 |                  | 89.86 %          | Converged         | Generative Model 1 | Aluminum AlSi10Mg | Unrestricted         | Group 2           |
|  Study 2 - Structu... - Outcome 1  |                  | 89.809 %         | Converged         | Generative Model 1 | Aluminum          | Unrestricted         | Group 2           |
|  Study 2 - Structu... - Outcome 15 |                  | 85.319 %         | Converged         | Generative Model 1 | ABS Plastic       | Casting              | Group 4           |
|  Study 2 - Structu... - Outcome 11 |                  | 75.49 %          | Converged         | Generative Model 1 | ABS Plastic       | Unrestricted         | Group 2           |
|  Study 2 - Structu... - Outcome 20 |                  | 69.851 %         | Converged         | Generative Model 1 | Aluminum AlSi10Mg | Additive             | Group 2           |
|  Study 2 - Structu... - Outcome 2  |                  | 69.8 %           | Converged         | Generative Model 1 | Aluminum          | Additive             | Group 2           |
|  Study 2 - Structu... - Outcome 17 |                  | 68.512 %         | Completed         | Generative Model 1 | Aluminum 6061     | 3 axis milling       | Group 1           |
|  Study 2 - Structu... - Outcome 4  |                  | 68.512 %         | Completed         | Generative Model 1 | Aluminum          | 3 axis milling       | Group 1           |
|  Study 2 - Structu... - Outcome 6  |                  | 68.163 %         | Converged         | Generative Model 1 | Steel             | Unrestricted         | Group 2           |
|  Study 2 - Structu... - Outcome 16 |                  | 67.137 %         | Converged         | Generative Model 1 | Aluminum 6061     | 2 axis cutting       | Group 3           |
|  Study 2 - Structu... - Outcome 3  |                  | 67.137 %         | Converged         | Generative Model 1 | Aluminum          | 2 axis cutting       | Group 3           |

Outcome filters

Sort by

Processing status

Recommended outcomes

Compare

 Study 2 - Structural ... - Outcome 1

Converged

 Study 2 - Structur... - Outcome 13

Converged

 Study 2 - Structur... - Outcome 15

Converged

 Study 2 - Structur... - Outcome 19

Converged

Converged

 Study 2 - Structural ... - Outcome 1

Converged

 Study 2 - Structural... - Outcome 2

Converged

 Study 2 - Structural... - Outcome 3

Converged

 Study 2 - Structural... - Outcome 5

Converged

## 10. Results

All 20 outcomes solved successfully (16 Converged, 4 Completed). They cover 20 of the 30 possible material–method combinations: six candidate materials were considered across five manufacturing methods, but the three aluminium alloys (AlSi10Mg, 6061 and A356-T6) were each explored on only a subset of the methods. Every outcome satisfies the Safety Factor  $\geq 2$  limit. Key results, sorted by mass:

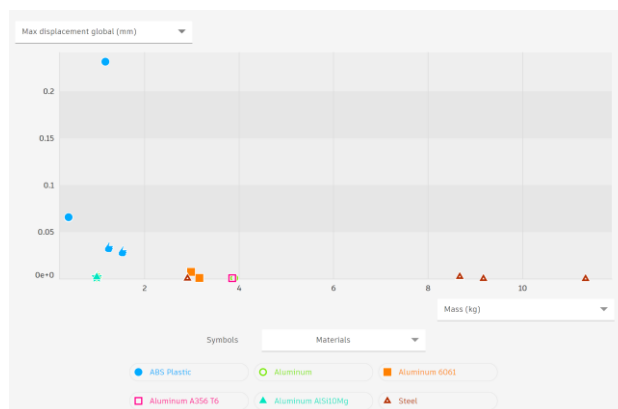
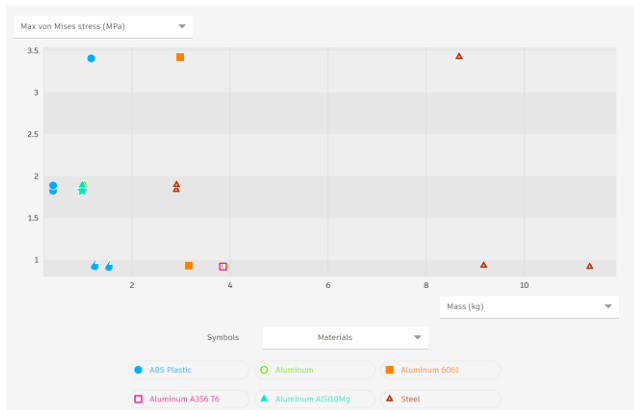
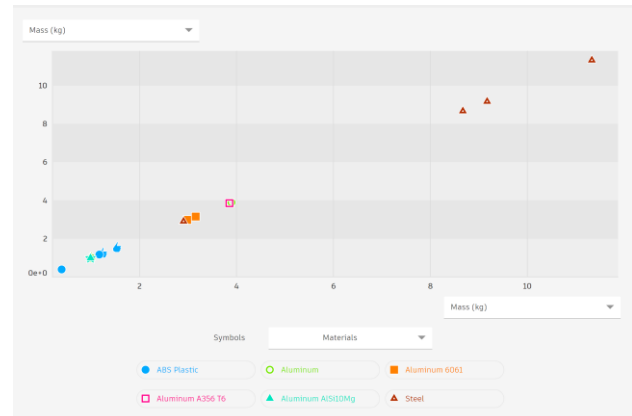
| Outcome   | Material                  | Process             | Mass (kg)    | Min SF       | $\sigma$ von Mises (MPa) | Max displ. (mm) | Rec. %      |
|-----------|---------------------------|---------------------|--------------|--------------|--------------------------|-----------------|-------------|
| 11        | ABS Plastic               | Unrestricted        | 0.392        | 10.95        | 1.83                     | 0.0653          | 75.5        |
| 12        | ABS Plastic               | Additive            | 0.393        | 10.58        | 1.89                     | 0.0659          | 55.2        |
| <b>19</b> | <b>Aluminium AlSi10Mg</b> | <b>Unrestricted</b> | <b>0.987</b> | <b>131.3</b> | <b>1.83</b>              | <b>0.0021</b>   | <b>89.9</b> |
| <b>20</b> | <b>Aluminium AlSi10Mg</b> | <b>Additive</b>     | <b>0.989</b> | <b>126.8</b> | <b>1.89</b>              | <b>0.0021</b>   | <b>69.9</b> |
| <b>1</b>  | <b>Aluminium</b>          | <b>Unrestricted</b> | <b>0.998</b> | <b>150.1</b> | <b>1.83</b>              | <b>0.0021</b>   | <b>89.8</b> |
| 2         | Aluminium                 | Additive            | 1.000        | 145.1        | 1.89                     | 0.0021          | 69.8        |
| 14        | ABS Plastic               | 3-axis milling      | 1.171        | 5.87         | 3.41                     | 0.2317          | 46.8        |
| 13        | ABS Plastic               | 2-axis cutting      | 1.240        | 21.48        | 0.93                     | 0.0333          | 90.0        |
| 15        | ABS Plastic               | Casting             | 1.531        | 21.68        | 0.92                     | 0.0289          | 85.3        |
| 6         | Steel                     | Unrestricted        | 2.903        | 112.6        | 1.84                     | 0.0007          | 68.2        |
| 7         | Steel                     | Additive            | 2.908        | 109.1        | 1.90                     | 0.0007          | 48.1        |
| 4 / 17    | Aluminium / 6061          | 3-axis milling      | 2.982        | 80.4         | 3.42                     | 0.0076          | 68.5        |
| 3 / 16    | Aluminium / 6061          | 2-axis cutting      | 3.159        | 294.9        | 0.93                     | 0.0011          | 67.1        |
| 18        | Aluminium A356-T6         | Casting             | 3.854        | 179.0        | 0.92                     | 0.0009          | 43.2        |
| 5         | Aluminium                 | Casting             | 3.897        | 298.3        | 0.92                     | 0.0009          | 41.8        |
| 9         | Steel                     | 3-axis milling      | 8.670        | 60.4         | 3.43                     | 0.0025          | 0.0         |
| 8         | Steel                     | 2-axis cutting      | 9.170        | 221.8        | 0.93                     | 0.0004          | 0.0         |
| 10        | Steel                     | Casting             | 11.330       | 225.2        | 0.92                     | 0.0003          | 0.0         |

\*(Outcomes 3/16 and 4/17 are identical geometries shared between generic Aluminium and Aluminium 6061.)

## 11. Selected solutions

The following diagrams present a graphical comparison of the most relevant outcomes generated by the study. Each point corresponds to an individual design solution and is identified by its material. The plots relate the principal engineering parameters, namely volume, mass, maximum von Mises stress, maximum global displacement and minimum safety factor, so that the trade-offs between weight reduction, structural performance and material selection can be assessed.

The star symbol (★) denotes the selected final solution (Outcome 19). This outcome was retained because it offers the most favourable overall balance between low mass, reduced volume, a high safety factor, low stress levels and adequate structural stiffness. Comparing its position with that of the remaining alternatives allows its relative advantages to be identified and the final design choice to be justified.



### 11.1. Outcome 19 – Selected Final Design

Outcome 19 was selected as the final design because it provides the most favourable overall compromise between weight reduction, structural performance and manufacturability. It records a mass of 0.987 kg, the lowest among the top-ranked solutions, together with a minimum safety factor of 131.33 and a maximum displacement of 0.00205 mm, which indicates a high structural stiffness under the applied load. Its recommendation score of 89.86 is among the highest returned by the software. Taken together, these values show that Outcome 19 combines a substantial reduction in component mass with mechanical performance well in excess of the design requirement.



Study 2 - Structu... - Outcome 19  
Iteration 26 (final)

| Properties                   |                    |
|------------------------------|--------------------|
| Status                       | Converged          |
| Generative model             | Generative Model 1 |
| Material                     | Aluminum AISI10Mg  |
| Orientation                  | -                  |
| Manufacturing method         | Unrestricted       |
| Visual similarity            | Group 2            |
| Volume (mm <sup>3</sup> )    | 369,472.207        |
| Mass (kg)                    | 0.986              |
| Max von Mises stress (MPa)   | 1.829              |
| Safety factor limit          | 2                  |
| Min safety factor            | 131.208            |
| Max displacement global (mm) | 0.002              |

### 11.2. Outcome 1 – Alternative High-Performance Solution



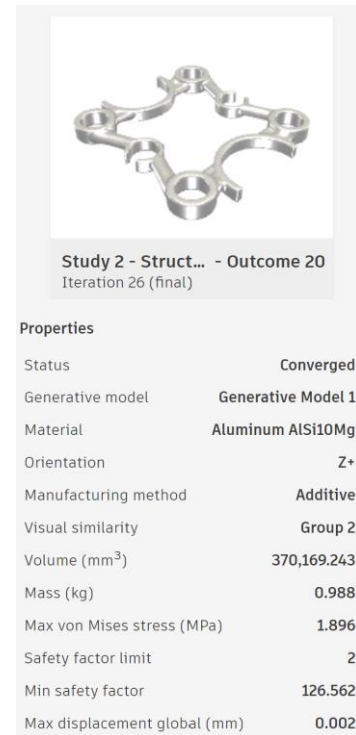
Study 2 - Structur... - Outcome 1  
Iteration 26 (final)

| Properties                   |                    |
|------------------------------|--------------------|
| Status                       | Converged          |
| Generative model             | Generative Model 1 |
| Material                     | Aluminum           |
| Orientation                  | -                  |
| Manufacturing method         | Unrestricted       |
| Visual similarity            | Group 2            |
| Volume (mm <sup>3</sup> )    | 369,470.449        |
| Mass (kg)                    | 0.998              |
| Max von Mises stress (MPa)   | 1.829              |
| Safety factor limit          | 2                  |
| Min safety factor            | 150.315            |
| Max displacement global (mm) | 0.002              |

Outcome 1 represents the closest alternative to the selected design, with a recommendation score of 89.81. It has a mass of 0.998 kg, only marginally greater than that of Outcome 19, and a higher minimum safety factor of 150.10. Its maximum displacement of 0.00211 mm again indicates high stiffness and structural stability. Although its overall performance is comparable to that of Outcome 19, the slightly higher mass excluded it from final selection. It nonetheless remains a strong candidate on account of its load-carrying capacity and reliability.

### 11.3. Outcome 20 – Best Additive Manufacturing Solution

Outcome 20 was retained among the leading candidates because it combines a high level of structural performance with a geometry generated specifically for additive manufacturing. It records a mass of 0.989 kg, a minimum safety factor of 126.76 and a maximum displacement of 0.00208 mm, values that are close to those of Outcomes 19 and 1. The solution is therefore well suited to additive production where that route is available. Although its recommendation score of 69.85 is lower than those of the other two solutions, its structural performance and manufacturability make it a viable alternative.



### 11.4 Final Ranking:

| Rank | Outcome    | Reason                                                                                                       |
|------|------------|--------------------------------------------------------------------------------------------------------------|
| 1    | Outcome 19 | Lowest mass among the metallic solutions, excellent stiffness, very high safety factor, best overall balance |
| 2    | Outcome 1  | Highest safety factor, excellent stiffness, slightly higher mass                                             |
| 3    | Outcome 20 | Excellent performance and optimized for additive manufacturing                                               |

### 11.5 Rejection of Outcome 13

Outcome 13 obtained one of the highest recommendation scores returned by the study. It was nevertheless rejected as the final solution once its overall engineering performance had been assessed against the project requirements.

The principal reason for rejection is the use of ABS plastic as the structural material. Although the design satisfies the minimum structural requirements, the original component is a load-bearing mechanical part that was analysed in steel; an aluminium-based solution therefore offers a more appropriate balance between structural performance and weight reduction.

In addition, Outcome 13 exhibits a maximum displacement of approximately 0.033 mm, appreciably greater than the value of about 0.002 mm obtained for the selected aluminium solutions. This reflects a lower stiffness and a larger deformation under the applied load.

Its mass of 1.24 kg is also higher than that of the selected solution (0.987 kg), so that, despite the favourable recommendation score, it does not represent the best available weight reduction.

On account of its higher mass, lower stiffness and the use of a polymer in place of a metallic structural material, Outcome 13 was therefore not retained. The final choice was Outcome 19, which provides a better compromise between mass reduction, structural stiffness, safety factor and manufacturability while maintaining a high level of mechanical performance.



Study 2 - Structu... - Outcome 13  
Iteration 39 (final)

| Properties                   |                    |
|------------------------------|--------------------|
| Status                       | Converged          |
| Generative model             | Generative Model 1 |
| Material                     | ABS Plastic        |
| Orientation                  | Z                  |
| Manufacturing method         | 2 axis cutting     |
| Visual similarity            | Group 3            |
| Volume (mm <sup>3</sup> )    | 1.17e+6            |
| Mass (kg)                    | 1.24               |
| Max von Mises stress (MPa)   | 0.928              |
| Safety factor limit          | 2                  |
| Min safety factor            | 21.558             |
| Max displacement global (mm) | 0.033              |

## 12. Conclusions

The generative design study applied the boundary conditions of the Project-3 static structural analysis to the same component and produced twenty admissible outcomes, each satisfying the imposed safety-factor limit of two. By varying the candidate material and the manufacturing method within a single study, the analysis made it possible to examine the influence of these choices on mass, stress, displacement and safety factor, and to position every solution against the original steel reference.

Although the two ABS-plastic outcomes were the lightest of all (about 0.39 kg), they were not retained: the component is a load-bearing structural part analysed in steel, for which a polymer is not an appropriate material. Among the metallic solutions the aluminium designs provide the best overall compromise. Outcome 19 (Aluminium AlSi10Mg, unrestricted) was therefore selected as the final design, with a mass of 0.987 kg — the lowest of the metallic outcomes — a minimum safety factor of 131.33 and a maximum displacement of 0.00205 mm. Outcome 1 (generic aluminium) and Outcome 20 (Aluminium AlSi10Mg, additive) were kept as the principal alternatives, the former offering a marginally higher safety factor and the latter a geometry suited to additive manufacturing. Relative to the steel reference the selected design reduces the mass by roughly two-thirds ( $\approx 2.9 \text{ kg} \rightarrow \approx 1.0 \text{ kg}$ ); this gain is achieved mainly through the change of material, since the very high safety factors show that the  $\text{SF} \geq 2$  limit is not the binding constraint under the light 1000 N load. Overall the study confirms generative design as an effective method for exploring material and manufacturing trade-offs in lightweight structural optimisation.